1 Two coils, P and Q, are placed close to each other, as shown in Fig. 1.1.

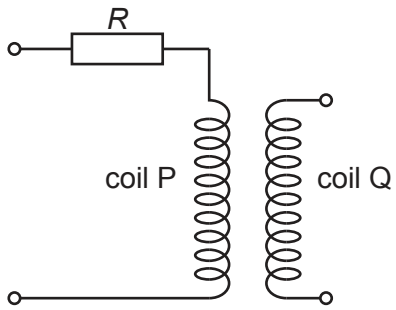

Fig. 1.1

A resistor of resistance R is connected in series with coil P.

A changing magnetic flux of frequency f in coil P causes an electromotive force (e.m.f.) E to be induced across the terminals of coil Q.

It is suggested that E is related to R by the relationship

$$E = 2\pi f M \left(\frac{V}{R + k}\right)$$

where V is the potential difference across the resistor and coil P, and k and M are constants.

Plan a laboratory experiment to test the relationship between E and R.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for k and M.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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